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POWER RAIL AND SIGNAL CONDUCTING LINE ARRANGEMENT

Abstract

An integrated circuit includes a first-voltage underlayer power rail and a second-voltage underlayer power rail extending in a first direction below a first connection layer. A first-type transistor and a second-type transistor are underneath the first connection layer. The source region of the first-type transistor is connected to the first-voltage underlayer power rail, and the source region of the second-type transistor is connected to the second-voltage underlayer power rail. The integrated circuit also includes a first-voltage power rail, a second-voltage power rail, and a signal conducting line, each of which extends in a second direction in the first connection layer. The first-voltage power rail is connected to the first-voltage underlayer power rail, and the second-voltage power rail is connected to the second-voltage underlayer power rail. The signal conducting line is conductively connected to either a terminal-conductor or a gate-conductor.

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Background/Summary

PRIORITY CLAIM [0001] The present application is a divisional of U.S. application Ser. No. 18/641,719, filed Apr. 22, 2024, now U.S. Pat. No. 12,317,589; issued May 27, 2025, which is a divisional of U.S. application Ser. No. 17/395,126, filed Aug. 5, 2021, now U.S. Pat. No. 11,967,596, issued Apr. 23, 2024, each of which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] The recent trend in miniaturizing integrated circuits (ICs) has resulted in smaller devices which consume less power yet provide more functionality at higher speeds. The miniaturization process has also resulted in stricter design and manufacturing specifications as well as reliability challenges. Various electronic design automation (EDA) tools generate, optimize and verify standard cell layout designs for integrated circuits while ensuring that the standard cell layout design and manufacturing specifications are met.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 is a layout diagram of an inverter circuit, in accordance with some embodiments.

[0005] FIGS. 2A-2C are cross-sectional view of the inverter circuit as specified by the layout diagram in FIG. 1, in accordance with some embodiments.

[0006] FIG. 3A is a layout diagram of an inverter circuit having a slot via-connector, in accordance with some embodiments.

[0007] FIG. 3B is a layout diagram of an inverter circuit having one power rail at each vertical boundary of the cell, in accordance with some embodiments.

[0008] FIG. 4A is a layout diagram of an And-Or-Invertor logic circuit having underlayer power rails, in accordance with some embodiments.

[0009] FIG. 4B is an equivalent circuit of the And-Or-Invertor logic circuit as specified by the layout diagram in FIG. 4A, in accordance with some embodiments.

[0010] FIG. 4C is a circuit diagram of the And-Or-Invertor logic circuit in FIG. 4A, in accordance with some embodiments.

[0011] FIG. 5A is a layout diagram of a transmission gate circuit having underlayer power rails, in accordance with some embodiments.

[0012] FIG. 5B is an equivalent circuit of the transmission gate circuit as specified by the layout diagram in FIG. 5A, in accordance with some embodiments.

[0013] FIG. 5C is a circuit diagram of the transmission gate circuit in FIG. 5A, in accordance with some embodiments.

[0014] FIG. 6 is a flow chart of a method of generating a layout design of an integrated circuit by a processor, in accordance with some embodiments.

[0015] FIGS. 7A-7E are partial layout diagrams of various circuits, in accordance with some embodiments.

[0016] FIG. 8 is a three-dimensional schematic drawing of the power distribution connections for an integrated circuit having underlayer power rails, in accordance with some embodiments.

[0017] FIGS. 9A-9B are schematic drawings of the power distribution connections between the power rails and the power stubs, in accordance with some embodiments.

[0018] FIG. 10 is a schematic drawing of a layout plan of power stubs in the second connection layer, in accordance with some embodiments.

[0019] FIG. 11 is a flow chart of a method of fabricating an integrated circuit having underlayer power rails, in accordance with some embodiments.

[0020] FIG. 12 is a block diagram of an electronic design automation (EDA) system in accordance with some embodiments.

[0021] FIG. 13 is a block diagram of an integrated circuit (IC) manufacturing system, and an IC manufacturing flow associated therewith, in accordance with some embodiments.

DETAILED DESCRIPTION

[0022] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components, values, operations, materials, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. Other components, values, operations, materials, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0023] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0024] An integrated circuit device generally includes various power rails for providing supply voltages to the transistors in the integrated circuit. The channel regions, the source regions, and the drain regions of the transistors are often aligned within the active zones extending in a first direction, while the gate-conductors intersecting the channel regions of the transistors generally extends in a second direction that is perpendicular to the first direction. The orientation of the signal conducting lines in a first connection layer (e.g., metal layer M0) immediately above the

gate-conductors may influence the cell widths of various cells in the integrated circuit devices. In some embodiments, if underlayer power rails extending in the first direction are provided below the first connection layer and if the signal conducting lines extending in the second direction are provided in the first connection layer, the cell widths of some cells in the integrated circuit devices are reduced as compared to the cells in alternative implementations with different signal conducting line orientations. In some embodiments, power rails extending in the second direction are also provided in the first connection layer. While the widths of the power rails in the first connection layer are restricted by the adjacent signal conducting lines in the same first connection layer, the widths of the underlayer power rails nevertheless are not restricted by the signal conducting lines in the first connection layer. Furthermore, despite the widths of the power rails in the first connection layer are restricted, when the power rails in the first connection layer extends in the second direction, it is possible to increase the number of the power rails in the first connection layer for some of the cells. Larger width of underlayer power rails and/or larger placement density of the power rails in the first connection layer reduce both electromigration effects and IR Drops in the integrated circuit devices, which improves the overall reliability of the IC devices.

[0025] FIG. 1 is a layout diagram of an inverter circuit **100**, in accordance with some embodiments. FIGS. 2A-2C are cross-sectional view of the inverter circuit as specified by the layout diagram in FIG. 1, in accordance with some embodiments. The layout diagram of FIG. 1 includes the layout patterns for specifying a p-type active zone **80p** and an n-type active zone **80n** both extending in the X-direction, a gate-conductor **115** extending in the Y-direction, terminal-conductors (**172p**, **172n**, and **178**) extending in the Y-direction, signal conducting lines **155** and **198** extending in the Y-direction, and various via-connectors. The layout diagram of FIG. 1 also includes the layout patterns for specifying a first-voltage underlayer power rail **40** and a second-voltage underlayer power rail **20** both extending in the X-direction, first-voltage power rails (**52** and **58**) and second-voltage power rails (**32** and **38**) all extending in the Y-direction.

[0026] In the inverter circuit **100** as specified by the layout diagram of FIG. 1, the gate-conductor **115** intersects a semiconductor structure in the p-type active zone **80p** at the channel region of a p-type transistor **T1p**. The gate-conductor **115** also intersects a semiconductor structure in the n-type active zone **80n** at the channel region of an n-type transistor **T1n**. The terminal-conductor **172p** intersects the semiconductor structure in the p-type active zone **80p** at the source region of the p-type transistor **T1p**. The terminal-conductor **172n** intersects the semiconductor structure in the n-type active zone **80n** at the source region of the n-type transistor **T1n**. The terminal-conductor **178** intersects the semiconductor structure in the p-type active zone **80p** at the drain region of the p-type transistor **T1p**, and intersects the semiconductor structure in the n-type active zone **80n** at the drain region of the n-type transistor **T1n**. In some embodiments, when the semiconductor structure in the p-type active zone **80p** and the semiconductor structure in the n-type active zone **80n** are fin structures, the p-type transistor **T1p** and the n-type transistor **T1n** are FinFETs. In some embodiments, when the semiconductor structure in the p-type active zone **80p** and the semiconductor structure in the n-type active zone **80n** are nano-sheet structures, the p-type transistor **T1p** and the n-type transistor **T1n** are nano-sheet transistors. In some embodiments, when the semiconductor structure in the p-type active zone **80p** and the semiconductor structure in the n-type active zone **80n** are nano-wire structures, the p-type transistor **T1p** and the n-type transistor **T1n** are nano-wire transistors. In FIG. 1, the dummy gate-conductor layout patterns **111** and **119** at the vertical boundaries of the inverter circuit **100** specify that the p-type active zone **80p** and the n-type active zone **80n** do not extend into adjacent cells. That is, none of the source regions, drain regions, and channel regions in the p-type active zone **80p** and the n-type active zone **80n** joints with other active regions (such as, source regions, drain regions, and channel regions) in the adjacent cells.

[0027] In the inverter circuit **100** as specified by the layout diagram of FIG. 1, the signal conducting lines **155** and **198** are positioned in a first connection layer **M0** (as shown in FIGS. 2A-

2C), which overlies the insulating material directly covering the gate-conductor **115** and the terminal-conductors (**172p**, **172n**, and **178**). The signal conducting line **155** is conductively connected to the gate-conductor **115** through the gate via-connector VG. The signal conducting line **198** is conductively connected to the terminal-conductor **178** through the terminal via-connector VD. The first-voltage power rails (**52**, **58**) and the second-voltage power rails (**32**, **38**) are in the first connection layer M0. The first-voltage underlayer power rail **40** and the second-voltage underlayer power rail **20** are below the first connection layer. The first-voltage underlayer power rail **40** is conductively connected to the first-voltage power rails **52** and **58** correspondingly through the via-connectors V52 and V58. The first-voltage underlayer power rail **40** is also conductively connected to the terminal-conductor **172p**. The second-voltage underlayer power rail **20** is conductively connected to the second-voltage power rails **32** and **38** correspondingly through the via-connectors V32 and V38. The second-voltage underlayer power rail **20** is also conductively connected to the terminal-conductor **172n**.

[0028] In some embodiments, as shown in FIG. 1, the conductive connection between the first-voltage underlayer power rail **40** and the terminal-conductor **172p** in an IC device is specified in the layout diagram by a common overlap region between the layout pattern for the first-voltage underlayer power rail **40** and the layout pattern for the terminal-conductor **172p**. In some alternative embodiments, a connection symbol (such as a via-connector pattern) is placed in the above described common overlap region to specify that the first-voltage underlayer power rail **40** is conductively connected to the terminal-conductor **172p**. Similarly, in some embodiments, as shown in FIG. 1, the conductive connection between the second-voltage underlayer power rail **20** and the terminal-conductor **172n** in an IC device is specified in the layout diagram by a common overlap region between the layout pattern for the second-voltage underlayer power rail **20** and the layout pattern for the terminal-conductor **172n**. In some alternative embodiments, a connection symbol (such as a via-connector pattern) is placed in the above described common overlap region in the layout diagram to specify that the second-voltage underlayer power rail **20** is conductively connected to the terminal-conductor **172n**. As a comparison, in the layout diagram of FIG. 1, because the layout pattern for the terminal-conductor **178** does not overlap with the layout patterns for the first-voltage underlayer power rail **40** and the second-voltage underlayer power rail **20**, the terminal-conductor **178** does not form direct conductive connections with the first-voltage underlayer power rail **40** and the second-voltage underlayer power rail **20** in the IC device as specified by the layout diagram.

[0029] FIG. 2A is a cross-sectional view of the inverter circuit **100** as specified by FIG. 1 in a cutting plane A-A', in accordance with some embodiments. In FIG. 2A, the terminal-conductor **172p** intersects the semiconductor structure **180p** (in the p-type active zone **80p**) at the source region sT1p of the p-type transistor T1p (which is identified in FIG. 1), and the terminal-conductor **172n** intersects the n-type first semiconductor structure **180n** (in the n-type active zone **80n**) at the source region sT1n of the n-type transistor T1n (which is identified in FIG. 1). The lower surface of the first-voltage underlayer power rail **40** is either at the same level as the upper surface of the terminal-conductor **172p** or at a level above the upper surface of the terminal-conductor **172p**. The second-voltage underlayer power rail **20** is either at the same level as the upper surface of the terminal-conductor **172n** or at a level above the upper surface of the terminal-conductor **172n**. The first-voltage underlayer power rail **40**, the second-voltage underlayer power rail **20**, and the terminal-conductors (**172p** and **172n**) are covered by the insulating material **210**.

[0030] In some embodiments, a portion of the first-voltage underlayer power rail **40** overlies directly on the upper surface of the terminal-conductor **172p** and forms a direct conductive contact with the terminal-conductor **172p**. In some alternative embodiments, the first-voltage underlayer power rail **40**, while above the terminal-conductor **172p**, is conductively connected to the terminal-conductor **172p** through an opening in the insulating material **210** between the first-voltage underlayer power rail **40** and the terminal-conductor **172p**. Similarly, in some embodiments, a

portion of the second-voltage underlayer power rail **20** overlies directly on the upper surface of the terminal-conductor **172n** and forms a direct conductive contact with the terminal-conductor **172n**. In some alternative embodiments, the second-voltage underlayer power rail **20**, while above the terminal-conductor **172n**, is conductively connected to the terminal-conductor **172n** through an opening in the insulating material **210** between the second-voltage underlayer power rail **20** and the terminal-conductor **172n**.

[0031] FIG. 2B is a cross-sectional view of the inverter circuit **100** as specified by FIG. 1 in a cutting plane B-B', in accordance with some embodiments. In FIG. 2B, the terminal-conductor **178** intersects the semiconductor structure **180p** (in the p-type active zone **80p**) at the drain region dT1p of the p-type transistor T1p (which is identified in FIG. 1) and intersects the semiconductor structure **180n** (in the n-type active zone **80n**) at the drain region dT1n of the n-type transistor T1n (which is identified in FIG. 1). In some embodiments, the lower surfaces of the first-voltage underlayer power rail **40** and the second-voltage underlayer power rail **20** are at the same level as the upper surface of the terminal-conductor **178**. In some embodiments, the lower surfaces of the first-voltage underlayer power rail **40** and the second-voltage underlayer power rail **20** are at a level above the upper surface of the terminal-conductor **178**. The first-voltage underlayer power rail **40**, the second-voltage underlayer power rail **20**, and the terminal-conductor **178** are covered by the insulating material **210**. The signal conducting line **198**, which is in the first connection layer M0, overlies the insulating material **210**. Both the first-voltage underlayer power rail **40** and the second-voltage underlayer power rail **20** are positioned below the first connection layer M0. The signal conducting line **198** is conductively connected to the terminal-conductor **178** through the terminal via-connector VD.

[0032] FIG. 2C is a cross-sectional view of the inverter circuit **100** as specified by FIG. 1 in a cutting plane C-C', in accordance with some embodiments. In FIG. 2C, the first-voltage power rail **58** and the second-voltage power rail **38**, which is in the first connection layer M0, overlies the insulating material **210**. The first-voltage power rail **58** is conductively connected to the first-voltage underlayer power rail **40** through the via-connector V58, and the second-voltage power rail **38** is conductively connected to the second-voltage underlayer power rail **20** through the via-connector V38. In operation, the first-voltage power rail **58** is configured to be maintained at a first supply voltage (such as a voltage VDD), and the second-voltage power rail **38** is configured to be maintained at a second supply voltage (such as a voltage VSS).

[0033] Various variations of the layout diagram in the FIG. 1 are depicted in FIGS. 3A-3B. FIG. 3A is a layout diagram of an inverter circuit **300** having a slot via-connector, in accordance with some embodiments. The layout diagram in the FIG. 3A is modified from the layout diagram in the FIG. 1. The modification includes adding a first-voltage power rail **55** and a second-voltage power rail **35**, replacing the signal conducting line **155** with the signal conducting line **192**, and replacing the gate via-connector VG with the slot gate via-connector VSG. The signal conducting line **192** is conductively connected to the gate-conductor **115** through the slot gate via-connector VSG. In FIG. 3A, because the slot gate via-connector VSG is extending in the X-direction, it becomes possible to connect the signal conducting line **192** to the gate-conductor **115**, even though the signal conducting line **192** and the gate-conductor **115** are shifted from each other along the X-direction.

[0034] In FIG. 3A, the first-voltage power rail **55** and the second-voltage power rail **35** are all positioned in the first connection layer M0. The first-voltage power rail **55** and the second-voltage power rail **35**, while aligned with the gate-conductor **115**, are separated from the first-voltage power rail **52** and the second-voltage power rail **32** along the X-direction by one Contacted Poly Pitch (CPP) distance. Here, one CPP distance is the pitch distance between two adjacent gate-conductors. The first-voltage power rail **55** is conductively connected to the first-voltage underlayer power rail **40** through the via-connector V55, and the second-voltage power rail **35** is conductively connected to the second-voltage underlayer power rail **20** through the via-connector V35. In some embodiments, when the first-voltage power rail **55** is separated from the first-voltage

power rail **52** by at least one CPP distance, the via-connector **V55** and the via-connector **V52** are able to be placed next to each other as shown in FIG. **3A** without violating the design rules. Similarly, when the second-voltage power rail **35** is separated from the second-voltage power rail **32** by at least one CPP distance, the via-connector **V35** and the via-connector **V32** are able to be placed next to each other as shown in FIG. **3A** without violating the design rules.

[0035] FIG. **3B** is a layout diagram of an inverter circuit **100** having one power rail at each vertical boundary of the cell, in accordance with some embodiments. The layout diagram in the FIG. **3B** is modified from the layout diagram in the FIG. **1**. The modification includes removing the second-voltage power rail **32**, the first-voltage power rail **58**, and the via-connectors **V32** and **V58**. The modification also includes replacing the first-voltage power rail **52** with the first-voltage power rail **352**, and replacing the second-voltage power rail **38** with the second-voltage power rail **338**.

[0036] In FIG. **3B**, the first-voltage power rail **352** is sufficiently extended in the Y-direction to completely cover the vertical boundary **301** of the cell containing the inverter circuit. The second-voltage power rail **338** is sufficiently extended in the Y-direction to completely cover the vertical boundary **309** of the cell containing the inverter circuit. Additionally, the first-voltage power rail **352** extends, along the vertical boundary **301**, from a first adjacent cell (which shares the horizontal boundary **302** with the inverter circuit **100** cell) to a second adjacent cell (which shares the horizontal boundary **308** with the inverter circuit **100** cell). The second-voltage power rail **338** extends, along the vertical boundary **309**, from the first adjacent cell to the second adjacent cell. The first-voltage power rail **352** is configured to provide the first supply voltage (such as VDD) to each of the first adjacent cell, the inverter circuit **100** cell, and the second adjacent cell. The second-voltage power rail **338** is configured to provide the second supply voltage (such as VSS) to each of the first adjacent cell, the inverter circuit **100** cell, and the second adjacent cell. In FIG. **3B**, the first-voltage power rail **352** is conductively connected to the first-voltage underlayer power rail **40** through the via-connector **V58**, and the second-voltage power rail **338** is conductively connected to the second-voltage underlayer power rail **20** through the via-connector **V38**.

[0037] FIG. **4A** is a layout diagram of an And-Or-Invertor (“AOI”) logic circuit having underlayer power rails, in accordance with some embodiments. FIG. **4B** is an equivalent circuit of the AOI logic circuit **400** as specified by the layout diagram in FIG. **4A**, in accordance with some embodiments. The AOI logic circuit **400** as specified by the layout diagram in FIG. **4A** is an example of the 2-2 AOI logic circuit in FIG. **4C**.

[0038] In FIG. **4C**, the source terminals of the p-type transistors having inputs **B1** and **B2** are connected to the power supply VDD. The drain terminals of the p-type transistors having inputs **B1** and **B2** are connected together as the node **n01**. The source terminals of the p-type transistors having inputs **A1** and **A2** are also connected to the node **n01**. The drain terminals of the p-type transistors having inputs **A1** and **A2** are connected together as the output terminal **ZN**. The drain terminals of the n-type transistors having inputs **A1** and **B1** are also connected to the output terminal **ZN**. The source terminal of the n-type transistor having inputs **A1** is connected to the drain terminal of the n-type transistor having input **A2**, and the source terminal of the n-type transistor having inputs **B1** is connected to the drain terminal of the n-type transistor having input **B2**. The source terminals of the n-type transistors having inputs **A2** and **B2** are connected to the power supply VSS.

[0039] The layout diagram of FIG. **4A** includes the layout patterns for specifying a p-type active zone **80p** and an n-type active zone **80n** both extending in the X-direction, the layout patterns for specifying a first-voltage underlayer power rail **40** and a second-voltage underlayer power rail **20** both extending in the X-direction, and the layout patterns for specifying the first-voltage power rails (**52**, **54**, and **58**) and the second-voltage power rails (**32** and **38**) all extending in the Y-direction. The layout diagram of FIG. **4A** also includes the layout patterns for specifying the gate-conductors (**gB2**, **gB1**, **gA1**, and **gA2**) extending in the Y-direction, the layout patterns for specifying the terminal-conductors (**472p**, **472n**, **474p**, **474n**, **475p**, **475n**, **476p**, **476n**, **478p**, and

478n) extending in the Y-direction, and the dummy gate-conductor layout patterns (**111** and **119**) extending in the Y-direction at the cell boundaries. The layout diagram of FIG. **4A** also includes the layout patterns for the signal conducting lines (**492**, **494**, **495p**, **495n**, **496**, and **498**) aligned with the corresponding terminal-conductors extending in the Y-direction, the layout patterns for the signal conducting lines (**454**, **456**, and **458**) extending in the Y-direction aligned the corresponding gate-conductors, the layout patterns for the horizontal signal conducting lines (**462** and **464**) extending in the X-direction, and the layout patterns for various via-connectors.

[0040] In the AOI logic circuit **400** as specified by the layout diagram in FIG. **4A**, each of the gate-conductors **gB2**, **gB1**, **gA1**, and **gA2**, extending in the Y-direction, intersects the p-type active zone **80p** and correspondingly forms the gate terminals of the p-type transistors **pB2**, **pB1**, **pA1**, and **pA2** (as identified in FIG. **4B**). Each of the gate-conductors **gB2**, **gB1**, **gA1**, and **gA2** also intersects the n-type active zone **80n** and correspondingly forms the gate terminals of the n-type transistors **nB2**, **nB1**, **nA1**, and **nA2** (as identified in FIG. **4B**). The terminal-conductors **472p**, **474p**, **475p**, **476p**, and **478p** overlap the p-type active zone **80p** and form the source/drain terminals of the p-type transistors. The terminal-conductors **472n**, **474n**, **475n**, **476n**, and **478n** overlap the n-type active zone **80n** and form the source/drain terminals of the n-type.

[0041] In the AOI logic circuit **400**, as shown in FIGS. **4A-4B**, the gate terminals of the p-type transistor **pB2** and the n-type transistor **nB2** are connected together by the gate-conductor **gB2**, the gate terminals of the p-type transistor **pB1** and the n-type transistor **nB1** are connected together by the gate-conductor **gB1**, the gate terminals of the p-type transistor **pA1** and the n-type transistor **nA1** are connected together by the gate-conductor **gA1**, and the gate terminals of the p-type transistor **pA2** and the n-type transistor **nA2** are connected together by the gate-conductor **gA2**. The gate-conductor **gB2** is conductively connected to the signal conducting line **494** through a slot gate via-connector **VSG**. Each of the gate-conductors **gB1**, **gA1**, and **gA2** is correspondingly connected to one of the signal conducting lines **454**, **456**, and **458** through a gate via-connector **VG**. Each of the signal conducting lines **494**, **454**, **456**, and **458** (correspondingly connected to the gate-conductors **gB2**, **gB1**, **gA1**, and **gA2**) is configured to receive one of the corresponding input signals **B2**, **B1**, **A1**, and **A2**.

[0042] In the AOI logic circuit **400**, as shown in FIGS. **4A-4B**, each of the terminal-conductors **472p**, **475p**, and **478p** is correspondingly connected to one of the signal conducting lines **492**, **495p**, and **498** through a terminal via-connector **VD**. Each of the signal conducting lines **492**, **495p**, and **498** is conductively connected to the horizontal signal conducting line **462** through a via-connector **VIA0**. Consequently, the drain terminals of the p-type transistors **pB2** and **pB1** are conductively connected to the source terminals of the p-type transistors **pA1** and **pA2** through the horizontal signal conducting line **462**. Furthermore, in the AOI logic circuit **400**, each of the terminal-conductors **476p** and **475n** is correspondingly connected to one of the signal conducting lines **496** and **495n** through a terminal via-connector **VD**. Each of the signal conducting lines **476** and **495n** is conductively connected to the horizontal signal conducting line **464** through a via-connector **VIA0**. Consequently, the drain terminals of the p-type transistors **pA1** and **pA2** are conductively connected to the drain terminals of the n-type transistors **nB1** and **nA1** through the horizontal signal conducting line **464** (which is configured to carry the output signal **ZN**). In the AOI logic circuit **400** as specified by the layout diagram in FIG. **4A**, the horizontal signal conducting lines **462** and **464** are in a second connection layer **M1**, which overlies the interlayer dielectric (ILD) materials separating the second connection layer **M1** and the first connection layer **M0**. In the AOI logic circuit **400** as specified by the layout diagram in FIG. **4A**, each of the via-connectors **VIA0** passes through the ILD materials separating the second connection layer **M1** and the first connection layer **M0**.

[0043] In the AOI logic circuit **400**, as shown in FIGS. **4A-4B**, the terminal-conductor **474p** is conductively connected to the first-voltage underlayer power rail **40** (which is also labeled as **VDD-VDR** in FIG. **4B**), and each of the terminal-conductors **472n** and **478n** is conductively

connected to the second-voltage underlayer power rail **20** (which is also labeled as VSS-VDR in FIG. **4B**). In the AOI logic circuit **400**, the first-voltage underlayer power rail **40** and the second-voltage underlayer power rail **20** are below the first connection layer **M0**. The first-voltage underlayer power rail **40** is conductively connected to the first-voltage power rails **52**, **54**, and **58** correspondingly through the via-connectors **V52**, **V54**, and **V58**, while the second-voltage underlayer power rail **20** is conductively connected to the second-voltage power rails **32**, **34**, and **38** correspondingly through the via-connectors **V32**, **V34**, and **V38**. In some embodiments, in the AOI logic circuit **400** of FIG. **4B**, when the first-voltage power rails **52**, **54**, and **58** are configured to receive the supply voltage VDD, the source terminals of the p-type transistors **pB2** and **pB1** are configured to be maintained at the supply voltage VDD. Similarly, in some embodiments, when the second-voltage power rails **32**, **34**, and **38** are configured to receive the supply voltage VSS, the source terminals of the n-type transistors **nB2** and **nA2** are configured to be maintained at the supply voltage VSS.

[0044] In the AOI logic circuit **400** as specified by the layout diagram in FIG. **4A**, the signal conducting lines in the first connection layer **M0** extend in a first direction which is the same extending direction of the gate-conductors, and the underlayer power rails (positioned below the first connection layer) extend in an orthogonal second direction. Due to the orientation of the signal conducting lines in the first connection layer **M0** and the orientation of the underlayer power rails, the AOI logic circuit **400** as specified in FIG. **4A** has a reduced cell width along the X-direction as compared with alternative implementations (in which the signal conducting lines in the first connection layer **M0** extend in an orthogonal direction crossing the gate-conductors). For example, the AOI logic circuit **400** as specified in FIG. **4A** has a cell width of 5 CPP, while alternative implementations with different signal conducting line orientations have a cell width of at least 6 CPP.

[0045] FIG. **5A** is a layout diagram of a transmission gate circuit **500** having underlayer power rails, in accordance with some embodiments. FIG. **5B** is an equivalent circuit of the transmission gate circuit **500** as specified by the layout diagram in FIG. **5A**, in accordance with some embodiments. The transmission gate circuit **500** as specified by the layout diagram in FIG. **5A** is an example of the transmission gate circuit in FIG. **5C**.

[0046] In FIG. **5C**, a transmission gate circuit includes two p-type transistors and two n-type transistors. One of the two p-type transistors has a gate “**A1**” and another one of the two p-type transistors has a gate “**B1**.” One of the two n-type transistors has a gate “**B2**” and another one of the two n-type transistors has a gate “**A1**.” The gate “**A1**” of the p-type transistor is connected to the gate “**A2**” of the n-type transistor. The gate “**B1**” of the p-type transistor is connected to the gate “**B2**” of the n-type transistor. Each of the two p-type transistors has one source/drain terminal joined together and connected as a joint node. Each of the two n-type transistors has one source/drain terminal joined together and connected to the joint node.

[0047] The layout diagram of FIG. **5A** includes the layout patterns for specifying a p-type active zone **80p** and an n-type active zone **80n** both extending in the X-direction, the layout patterns for specifying a first-voltage underlayer power rail **40** and a second-voltage underlayer power rail **20** both extending in the X-direction, and the layout patterns for specifying the first-voltage power rail **52** and the second-voltage power rail **38** both extending in the Y-direction. The layout diagram of FIG. **5A** also includes the layout patterns for specifying the gate-conductors (**gA1**, **gB1**, **gB2**, and **gA2**) extending in the Y-direction, and the layout patterns for specifying the terminal-conductors (**572p**, **572n**, **475**, **578p**, and **578n**) extending in the Y-direction, and the dummy gate-conductor layout patterns (**111** and **119**) extending in the Y-direction at the cell boundaries. The layout diagram of FIG. **5A** also includes the layout patterns for the signal conducting lines (**551**, **552p**, **552n**, **558p**, **558n**, and **559**) extending in the Y-direction, the layout patterns for the horizontal signal routing lines (**562**, **564**, **566** and **568**) extending in the X-direction, and the layout patterns for various via-connectors.

[0048] In the transmission gate circuit **500** as specified by the layout diagram in FIG. 5A, the signal conducting lines (**551**, **552_p**, **552_n**, **558_p**, **558_n**, and **559**) extending in the Y-direction are in the first connection layer **M0**. The horizontal signal routing lines (**562**, **564**, **566** and **568**) extending in the X-direction are in the second connection layer **M1**, which overlies the interlayer dielectric (ILD) materials separating the second connection layer **M1** and the first connection layer **M0**.

[0049] In the transmission gate circuit **500** as specified by the layout diagram in FIG. 5A, each of the gate-conductors **gA1** and **gB1**, extending in the Y-direction, intersects the p-type active zone **80_p** and correspondingly forms the gate terminals of the p-type transistors **pA1** and **pB1** (as identified in FIG. 5B). Each of the gate-conductors **gB2** and **gA2**, extending in the Y-direction, intersects the n-type active zone **80_n** and correspondingly forms the gate terminals of the n-type transistors **nB2** and **nA2** (as identified in FIG. 5B). The terminal-conductors **572_p**, **575**, and **578_p** intersect the p-type active zone **80_p** and form the source/drain terminals of the p-type transistors **pA1** and **pB1**. The terminal-conductors **572_n**, **575**, and **578_n** intersect the n-type active zone **80_n** and form the source/drain terminals of the n-type transistors **nB2** and **nA2**.

[0050] In the transmission gate circuit **500**, as shown in FIGS. 5A-5B, the gate-conductor **gA1** is conductively connected to the signal conducting line **552_p** through a corresponding gate via-connector **VG**, the signal conducting line **552_p** is conductively connected to the horizontal signal routing lines **564** through a corresponding gate via-connector **VIA0**, and the horizontal signal routing lines **564** is conductively connected to the signal conducting line **551** through a corresponding gate via-connector **VIA0**. In addition, the signal conducting line **551** is conductively connected to the horizontal signal conducting lines **568** through a corresponding gate via-connector **VIA0**, the horizontal signal conducting lines **568** is conductively connected to the signal conducting line **558_n** through a corresponding gate via-connector **VIA0**, and the signal conducting line **558_n** is conductively connected to the gate-conductor **gA2** through a corresponding gate via-connector **VG**. Consequently, the gate-conductor **gA1** is conductively connected to the gate-conductor **gA2**.

[0051] Similarly, as shown in FIGS. 5A-5B, the gate-conductor **gB1** is conductively connected to the signal conducting line **558_p** through a corresponding gate via-connector **VG**, the signal conducting line **558_p** is conductively connected to the horizontal signal routing lines **562** through a corresponding gate via-connector **VIA0**, and the horizontal signal routing lines **562** is conductively connected to the signal conducting line **559** through a corresponding gate via-connector **VIA0**. In addition, the signal conducting line **559** is conductively connected to the horizontal signal routing lines **566** through a corresponding gate via-connector **VIA0**, the horizontal signal routing lines **566** is conductively connected to the signal conducting line **552_n** through a corresponding gate via-connector **VIA0**, and the signal conducting line **552_n** is conductively connected to the gate-conductor **gB2** through a corresponding gate via-connector **VG**. Consequently, the gate-conductor **gB1** is conductively connected to the gate-conductor **gB2**.

[0052] In the AOI logic circuit **400** as specified by the layout diagram in FIG. 4A, the signal conducting lines in the first connection layer **M0** extend in a first direction which is the same extending direction of the gate-conductors, and the underlayer power rails (positioned below the first connection layer) extend in an orthogonal second direction. Due to the orientation of the signal conducting lines in the first connection layer **M0** and the orientation of the underlayer power rails, the AOI logic circuit **400** as specified in FIG. 4A has a reduced cell width along the X-direction as compared with alternative implementations (in which the signal conducting lines in the first connection layer **M0** extend in an orthogonal direction crossing the gate-conductors). For example, the AOI logic circuit **400** as specified in FIG. 4A has a cell width of 5 CPP, while alternative implementations with different signal conducting line orientations have a cell width of at least 6 CPP.

[0053] In the transmission gate circuit **500** as specified by the layout diagram in FIG. 5A, the signal conducting lines in the first connection layer **M0** extend in a first direction which is the same

extending direction of the gate-conductors, and the underlayer power rails (positioned below the first connection layer) extend in an orthogonal second direction. Due to the orientation of the signal conducting lines in the first connection layer **M0** and the orientation of the underlayer power rails, the signal connections in the transmission gate circuit **500** as specified in FIG. 5A are conducted by the signal conducting lines in the first connection layer **M0** and the horizontal signal conducting lines in the second connection layer **M1**. As a comparison, in some alternative implementations (in which the signal conducting lines in the first connection layer **M0** extends in an orthogonal direction crossing the gate-conductors), a different layout of the transmission gate circuit in FIG. 5C require a routing line in the third connection layer **M2** to complete the signal connections. The cell width of the transmission gate circuit **500** in some alternative implementations is larger than the cell width of the transmission gate circuit **500** as specified in FIG. 5A. The third connection layer **M2** (which is above the second connection layer **M1**) overlies the interlayer dielectric (ILD) materials separating the third connection layer **M2** and the second connection layer **M1**.

[0054] FIG. 6 is a flow chart of a method **600** of generating a layout design of an integrated circuit by a processor, in accordance with some embodiments. It is understood that additional operations may be performed before, during, and/or after the method **600** depicted in FIG. 6, and that some other processes may only be briefly described herein. In some embodiments, the method **600** is usable to generate one or more layout designs, such as the layout designs in FIG. 1, FIGS. 3A-3B, FIG. 4A, or FIG. 5A. In some embodiments, the method **600** is usable to form integrated circuits having similar structural relationships as among one or more of the elements in FIG. 1, FIGS. 2A-2C, FIGS. 3A-3B, FIG. 4A, or FIG. 5A. In some embodiments, method **600** is performed by a processing device (e.g., processor **1202** in FIG. 12) configured to execute instructions for generating one or more layout designs, such as the layout designs in FIG. 1, FIGS. 3A-3B, FIG. 4A, or FIG. 5A.

[0055] In operation **610** of method **600**, underlayer power rail patterns extending in a first direction are generated. The underlayer power rail patterns generated include a first-voltage underlayer power rail pattern and a second-voltage underlayer power rail pattern. In the example layout designs of in FIG. 1, FIGS. 3A-3B, FIG. 4A, and FIG. 5A, the first-voltage underlayer power rail **40** extending in the X-direction is specified by a corresponding first-voltage underlayer power rail pattern, and the second-voltage underlayer power rail **20** extending in the X-direction is specified by a corresponding second-voltage underlayer power rail pattern. Both the first-voltage underlayer power rail **40** and the second-voltage underlayer power rail **20** are positioned below the first connection layer **M0**. After operation **610**, the flow proceeds to operation **620**.

[0056] In operation **620** of method **600**, power rail patterns extending in a second direction are generated for specifying the power rails in the first connection layer. The power rail patterns generated include at least one first-voltage rail pattern and at least one second-voltage power rail pattern. In the example layout designs of FIG. 1, FIGS. 3A-3B, and FIG. 4A, each of the first-voltage power rails (such as, **52** and **58**, and also **55** in FIG. 3A and **54** in FIG. 4A) extending in the Y-direction is specified by a corresponding first-voltage power rail pattern, and the second-voltage power rails (such as, **32** and **38**, and also **35** in FIG. 3A and **34** in FIG. 4A) extending in the Y-direction is specified by a corresponding second-voltage power rail pattern. Each of the first-voltage power rails and the second-voltage power rails are in the first connection layer **M0**. After operation **620**, the flow proceeds to operation **625**.

[0057] In operation **625** of method **600**, via-connector patterns are positioned such that each via-connector pattern specifies the connection between a power rail in the first connection layer and an underlayer power rail. In the example layout designs of FIG. 1, FIGS. 3A-3B, and FIG. 4A, each connection between the first-voltage underlayer power rail **40** and one of the first-voltage power rails (such as, **52** and **58**, and also **55** in FIGS. 3A-3B and **54** in FIG. 4A) is specified by a corresponding via-connector pattern (such as, **V52** and **V58**, and also **V55** in FIG. 3A and **V54** in FIG. 4A). In the example layout designs of FIG. 1, FIGS. 3A-3B, and FIG. 4A, each connection

between the second-voltage underlayer power rail **20** and one of the second-voltage power rails (such as, **32** and **38**, and also **35** in FIG. 3A and **34** in FIG. 4A) is specified by a corresponding via-connector pattern (such as, V32 and V38, and also V35 in FIG. 3A and V34 in FIG. 4A). After operation **625**, the flow proceeds to operation **630**.

[0058] In operation **630** of method **600**, active zone patterns extending in the first direction are generated. The active zone patterns generated include a first-type active zone pattern and a second-type active zone pattern. In the example layout designs of in FIG. 1, FIGS. 3A-3B, FIG. 4A, and FIG. 5A, the p-type active zone **80p** extending in the X-direction is specified by a corresponding p-type active zone pattern, and the n-type active zone **80n** extending in the X-direction is specified by a corresponding n-type active zone pattern. After operation **630**, the flow proceeds to operation **640**.

[0059] In operation **640** of method **600**, terminal-conductor patterns are generated such that each terminal-conductor pattern intersects one underlayer power rail pattern and one active zone pattern. The terminal-conductor patterns generated includes a first terminal-conductor pattern intersecting the first-type active zone pattern and the first-voltage underlayer power rail pattern. The terminal-conductor patterns generated also includes a second terminal-conductor pattern intersecting the second-type active zone pattern and the second-voltage underlayer power rail pattern. For example, in the layout designs of FIG. 1 and FIGS. 3A-3B, the terminal-conductor pattern intersecting the layout pattern for the first-voltage underlayer power rail **40** and the layout pattern for the p-type active zone specifies the conductive connection between the terminal-conductor **172p** and the first-voltage underlayer power rail **40**. Also in the layout design of in FIG. 1 and FIGS. 3A-3B, the terminal-conductor pattern intersecting the layout pattern for the second-voltage underlayer power rail **20** and the layout pattern for the n-type active zone specifies the conductive connection between the terminal-conductor **172n** and the second-voltage underlayer power rail **20**. In some embodiments, additional terminal-conductor patterns are also generated before, during, or after operation **640**. For example, in the layout designs of FIG. 1 and FIGS. 3A-3B, the additional terminal-conductor patterns include the layout pattern for terminal-conductor **178**. After operation **640**, the flow proceeds to operation **650**.

[0060] In operation **650** of method **600**, gate-conductor patterns extending in the second direction are generated. In the example layout designs of FIG. 1 and FIGS. 3A-3B, the gate-conductor **115** is specified by a corresponding gate-conductor pattern extending in the Y-direction. In the example layout design of FIG. 4A, each of the gate-conductors gB2, gB1, gA1, and gA2 is specified by a corresponding gate-conductor pattern extending in the Y-direction. In the example layout design of FIG. 5A, each of the gate-conductors gA1, gB1, gB2, and gA2 is specified by a corresponding gate-conductor pattern extending in the Y-direction. After operation **650**, the flow proceeds to operation **660**.

[0061] In operation **660** of method **600**, signal conducting line patterns extending in the second direction are generated. In the example layout designs of FIG. 1 and FIG. 3A, the signal conducting line **155** is specified by a corresponding signal conducting line pattern extending in the Y-direction. In the example layout design of FIG. 3B, the signal conducting line **192** is specified by a corresponding signal conducting line pattern extending in the Y-direction. In the example layout designs of FIG. 4A, each of the signal conducting lines (**492**, **494**, **495p**, **495n**, **496**, **498**, **454**, **456**, and **458**) is specified by a corresponding signal conducting line pattern extending in the Y-direction. In the example layout designs of FIG. 5A, each of the signal conducting lines (**551**, **552p**, **552n**, **558p**, **558n**, and **559**) is specified by a corresponding signal conducting line pattern extending in the Y-direction. After operation **650**, the flow proceeds to operation **665**.

[0062] In operation **665** of method **600**, gate via-connector patterns are positioned such that each gate via-connector pattern specifies the connection between a gate-conductors and a signal conducting line. In the example layout designs of FIG. 1 and FIG. 3B, the gate via-connector pattern is generated to specify the gate via-connector VG for connecting the signal conducting line

155 with the gate-conductor **115**. In the example layout design of FIG. **3A**, the gate via-connector pattern is generated to specify the slot gate via-connector VSG for connecting the signal conducting line **195** with the gate-conductor **115**. In the example layout design of FIG. **4A**, the connections specified by the gate via-connector patterns include the connection between the gate-conductor gB2 and the signal conducting line **494**, the connection between the gate-conductor gB1 and the signal conducting line **454**, the connection between the gate-conductor gA1 and the signal conducting line **456**, and the connection between the gate-conductor gA2 and the signal conducting line **458**. In the example layout design of FIG. **5A**, the connections specified by the gate via-connector patterns include the connection between the gate-conductor gA1 and the signal conducting line **552p**, the connection between the gate-conductor gB1 and the signal conducting line **558p**, the connection between the gate-conductor gB2 and the signal conducting line **552n**, and the connection between the gate-conductor gA2 and the signal conducting line **558n**. After operation **650**, the flow proceeds to operation **670**.

[0063] In operation **670** of method **600**, power stub patterns extending in the first direction are generated. In some embodiments, each power stub pattern generated specifies a power stub in a second connection layer (such as the M1 layer) that is above the first connection layer (such as the M0 layer). After operation **670**, in operation **675** of method **600**, VIA0 via-connector patterns are positioned such that each VIA0 via-connector pattern specifies the connection between a power stub and a power rail in the first connection layer.

[0064] In some embodiments, in the example circuit cells as specified by the layout diagrams in FIG. **1**, FIGS. **3A-3B**, FIG. **4A**, and FIG. **5A**, each power rail is associated with at least one power stub. FIG. **7A** is a partial layout diagram of the inverter circuit **100** in FIG. **1**, in accordance with some embodiments. In FIG. **7A**, for clarity of the presentation, only the power rail patterns and the underlayer power rail patterns in FIG. **1** are retained along with the associated via-connector patterns, and additionally, power stub patterns are included. As specified by the partial layout diagram of FIG. **7A**, the first-voltage power rails **52** and **58** are correspondingly connected to the power stubs **S52** and **S58** through the VIA0 via-connectors at the corresponding intersections, and the second-voltage power rails **32** and **38** are also correspondingly connected to the power stubs **S32** and **S38** through the VIA0 via-connectors at the corresponding intersections.

[0065] FIG. **7B** is a partial layout diagram of the inverter circuit **300** in FIG. **3A**, in accordance with some embodiments. In FIG. **7B**, for clarity of presentation, only the power rail patterns and the underlayer power rail patterns in FIG. **3A** are retained along with the associated via-connector patterns, and additionally, power stub patterns are included. As specified by the partial layout diagram of FIG. **7B**, the first-voltage power rails (**52**, **55**, and **58**) are correspondingly connected to the power stubs (**S52**, **S55**, and **S58**) through the VIA0 via-connectors at the corresponding intersections, and the second-voltage power rails (**32**, **35**, and **38**) are also correspondingly connected to the power stubs (**S32**, **S35**, and **S38**) through the VIA0 via-connectors at the corresponding intersections.

[0066] FIG. **7C** is a partial layout diagram of the inverter circuit **100B** in FIG. **3B**, in accordance with some embodiments. In FIG. **7C**, for clarity of the presentation, only the power rail patterns and the underlayer power rail patterns in FIG. **3B** are retained along with the associated via-connector patterns, and additionally, power stub patterns are included. As specified by the partial layout diagram of FIG. **7C**, the first-voltage power rail **352** is connected to the power stubs **S52** and **S352** through the VIA0 via-connectors at the corresponding intersection, and the second-voltage power rail **332** is connected to the power stubs **S32** and **S332** through the VIA0 via-connectors at the corresponding intersection.

[0067] FIG. **7D** is a partial layout diagram of the AOI logic circuit **400** in FIG. **4A**, in accordance with some embodiments. In FIG. **7D**, for clarity of presentation, only the power rail patterns and the underlayer power rail patterns in FIG. **4A** are retained along with the associated via-connector patterns, and additionally, power stub patterns are included. As specified by the partial layout

diagram of FIG. 7D, the first-voltage power rails (52, 55, and 58) are correspondingly connected to the power stubs (S52, S54, and S58) through the VIA0 via-connectors at the corresponding intersections, and the second-voltage power rails (32, 35, and 38) are also correspondingly connected to the power stubs (S32, S34, and S38) through the VIA0 via-connectors at the corresponding intersections.

[0068] FIG. 7E is a partial layout diagram of the transmission gate circuit 500 in FIG. 5A, in accordance with some embodiments. In FIG. 7E, for clarity of the presentation, only the power rail patterns and the underlayer power rail patterns in FIG. 5A are retained along with the associated via-connector patterns, and additionally, power stub patterns are included. As specified by the partial layout diagram of FIG. 7E, the first-voltage power rails 52 is connected to the power stub S52 through the VIA0 via-connector at the corresponding intersection, and the second-voltage power rail 38 is also connected to the power stub S38 through the VIA0 via-connector at the corresponding intersection.

[0069] FIG. 8 is a three-dimensional schematic drawing of the power distribution connections for an integrated circuit having underlayer power rails, in accordance with some embodiments. In FIG. 8, the first supply voltage (such as VDD) for the integrated circuit is provided by the first-voltage underlayer power rail (such as 40A), and the second supply voltage (such as VSS) for the integrated circuit is provided by the second-voltage underlayer power rail (such as 20A or 20B). For example, the second-voltage underlayer power rail 20A crosses over an array of terminal-conductors 870, and some of the terminal-conductors (while functioning as the source terminals of the n-type transistors) are conductively connected to the second-voltage underlayer power rail 20A through the terminal via-connectors VD. In FIG. 8, the first-voltage underlayer power rail 40A is conductively connected to the first-voltage power rails 852 and 854 through the corresponding via-connectors under the first-voltage power rails, and each of the second-voltage underlayer power rails 20A and 20B is conductively connected to the second-voltage power rail 832 through the corresponding via-connector under the second-voltage power rail. The first-voltage power rails 852 and 854 and the second-voltage power rail 832 are all in the first connection layer M0.

[0070] In FIG. 8, each of the first-voltage power rails 852 and 854 is conductively connected to a corresponding one of the first-voltage power stubs S852 and S854 through the via-connector VIA0, and the second-voltage power rail 832 is conductively connected to the power stub S832 through the via-connector VIA0. Each of the power stubs S852, S832 and S854 extends in the X-direction and is in the second connection layer M1.

[0071] In FIG. 8, each of the first-voltage power stubs S852 and S854 is conductively connected to a corresponding one of the first-voltage power stubs P852 and P854 (extending in the Y-direction) through the via-connector VIA1, and the second-voltage power stub S832 is conductively connected to the second-voltage power stub P832 (extending in the Y-direction) through the via-connector VIA1. The power stubs P852, P832 and P854 are in the third connection layer M2 which is above in the second connection layer M1.

[0072] In FIG. 8, each of the first-voltage power stubs P852 and P854 in the third connection layer is conductively connected to a corresponding one of the first-voltage power stubs Q852 and Q854 (extending in the X-direction) through the via-connector VIA2, and the second-voltage power stub P832 in the third connection layer is conductively connected to the second-voltage power stub Q832 (extending in the X-direction) through the via-connector VIA2. The power stubs Q852, Q832 and Q854 are in the fourth connection layer M3 which is above the third connection layer M2.

[0073] In FIG. 8, each of the first-voltage power stubs Q852 and Q854 in the fourth connection layer is conductively connected to a corresponding one of the fifth-connection-layer power rails 852M4 and 854M4 (extending in the Y-direction) through the via-connector VIA3, and the second-voltage power stub Q832 in the fourth connection layer is conductively connected to the fifth-connection-layer power rails 832M4 (extending in the Y-direction) through the via-connector VIA3. The fifth-connection-layer power rails 852M4, 832M4, and 854M4 are in the fifth

connection layer M4 which is above in the fourth connection layer M3.

[0074] In FIG. 8, each of the fifth-connection-layer power rail **852M4** and **854M4** is conductively connected to the sixth-connection-layer power rail **840M5** (extending in the X-direction) through the via-connector VIA4, and the sixth-connection-layer power rail **840M5** is conductively connected to the seventh-connection-layer power rail **850M6** (extending in the Y-direction) through the via-connector VIA5. The fifth-connection-layer power rail **832M4** is conductively connected to the sixth-connection-layer power rail **820M5** (extending in the X-direction) through the via-connector VIA4, and the sixth-connection-layer power rail **820M5** is conductively connected to the seventh-connection-layer power rail **830M6** (extending in the Y-direction) through the via-connector VIA5.

[0075] FIGS. 9A-9B are schematic drawings of the power distribution connections between the power rails in the first connection layer and the power stubs in the second connection layer, in accordance with some embodiments. In FIGS. 9A-9B, the first-voltage power rails **952**, **954**, **956**, and **958** in the first connection layer M0 are all extending in the Y-direction, and the second-voltage power rails **932**, **934**, and **936** in the first connection layer M0 are all extending in the X-direction. The pitch distance along the X-direction between two first-voltage power rails is 8.0 CPP, and the pitch distance along the X-direction between two second-voltage power rails is also 8.0 CPP. The first-voltage power rails and the second-voltage power rails are positioned near the boundaries of the cells. For example, in some embodiments, each of the first-voltage power rails and the second-voltage power rails has a center line that is on a vertical boundary or near the vertical boundary (or displaced along the X-direction from the vertical boundary with a displacement less than 0.25 CPP, for the purpose of interpreting the meaning of near the vertical boundary).

[0076] In FIG. 9A, each first-voltage power rail (**952**, **954**, **956**, or **958**) is conductively connected to at least two corresponding first-voltage power stubs (S952A-S952B, S954A-S954B, S956A-S956B, or S958A-S958B) at the corresponding intersections, and each second-voltage power rail (**932**, **934**, or **936**) is conductively connected to at least two corresponding second-voltage power stubs (S932A-S932B, S934A-S934B, or S936A-S936B) at the corresponding intersections. In FIG. 9B, each first-voltage power rail (**952**, **954**, **956**, or **958**) is conductively connected to at least three corresponding first-voltage power stubs (S952A-S952C, S954A-S954C, S956A-S956C, or S958A-S958C) at the corresponding intersections, and each second-voltage power rail (**932**, **934**, or **936**) is conductively connected to at least three corresponding second-voltage power stubs (S932A-S932C, S934A-S934C, or S936A-S936C) at the corresponding intersections. In FIGS. 9A-9B, the first-voltage power stubs and the second-voltage power stubs are in the second connection layer M1. Each of the first-voltage power stubs and the second-voltage power stubs extends in the X-direction and has a length in a range from 2.0 CPP to 3.0 CPP. Each row of the first-voltage power stubs (e.g., **952B**) is displaced along the Y-direction from another row of the second-voltage power stubs (e.g., **932B**), and the displacement distance along the Y-direction between the two rows is larger than or equal to the minimal pitch distance between two metal lines in the second connection layer M1. The first-voltage power stubs and the second-voltage power stubs are near the boundaries of the cells. For example, in some embodiments, each of the first-voltage power stubs and the second-voltage power stubs has a center point on a vertical cell boundary or near the vertical cell boundary (or displaced along the X-direction from the vertical cell boundary with a displacement less than 0.25 CPP, for the purpose of interpreting the meaning of near the vertical cell boundary).

[0077] In FIG. 9A, the pitch distance along the Y-direction between two first-voltage power stubs (e.g., between S952A and S952B) crossing a same first-voltage power rail (e.g., **952**) is four times the cell height, and the pitch distance along the Y-direction between two second-voltage power stubs (e.g., between S932A and S932B) crossing a same second-voltage power rail (e.g., **952**) is also four times the cell height. In FIG. 9B the pitch distance along the Y-direction between two first-voltage power stubs (e.g., between S952A and S952B, or between S952B and S952C) crossing a same first-voltage power rail (e.g., **952**) is two times the cell height, and the pitch distance along

the Y-direction between two second-voltage power stubs (e.g., between S932A and S932B, or between S932B and S932C) crossing a same second-voltage power rail (e.g., 952) is also two times the cell height.

[0078] FIG. 10 is a schematic drawing of a layout plan of power stubs in the second connection layer, in accordance with some embodiments. In FIG. 10, the power stub S1038 intersects a vertical boundary of a cell 1010, and the power stub S1052 intersects a vertical boundary of a cell 1020. In some embodiments, power stubs in the second connection layer M1 are near the horizontal cell boundaries. For example, in FIG. 10, the power stub S1038 is aligned in the same signal track as the output pin 1068 adjacent to the horizontal boundary 1012 of the cell 1010, and the power stub S1052 is aligned with an empty track 1062 adjacent to the horizontal boundary 1022 of the cell 1020. Additionally, each pin (such as the output pin 1068) sharing a same signal track with a power stub (such as the power stub S1038) generally has at least 1.5 CPP separation distance between the pin and the nearest vertical cell boundary (such as the boundary 1019).

[0079] FIG. 11 is a flow chart of a method 1100 of fabricating an integrated circuit having underlayer power rails, in accordance with some embodiments. It is understood that additional operations may be performed before, during, and/or after the method 1100 depicted in FIG. 11. In some embodiments, the method 1100 is usable in the manufacture processes for fabricating one or more integrated circuits, such as the integrated circuits as specified by the layout designs in FIG. 1, FIGS. 3A-3B, FIG. 4A, or FIG. 5A. In some embodiments, the method 1100 is usable in the manufacture processes for fabricating one or more integrated circuits having similar structural relationships as among one or more of the elements in FIG. 1, FIGS. 2A-2C, FIGS. 3A-3B, FIG. 4A, or FIG. 5A.

[0080] In operation 1110 of method 1100, the first semiconductor structure and the second semiconductor structure are fabricated. In some embodiments, the first semiconductor structure is fabricated for supporting one or more p-type transistors, and the second semiconductor structure is fabricated for supporting one or more n-type transistors. In some embodiments, the first semiconductor structure is fabricated for supporting one or more n-type transistors, and the second semiconductor structure is fabricated for supporting one or more p-type transistors. In some embodiments, the semiconductor structures are fabricated as fin structures. In some embodiments, the semiconductor structures are fabricated as nano-sheet structures. In some embodiments, the semiconductor structures are fabricated as nano-wire structures. As a non-limiting example, in FIGS. 2A-2C, the semiconductor structure 180p for supporting the p-type transistor T1p (which is identified in FIG. 1) and the semiconductor structure 180n for supporting the n-type transistor T1n (which is identified in FIG. 1) are fabricated as nano-sheet structures.

[0081] In operation 1120 of method 1100, first-type transistors and second-type transistors are fabricated. The fabrication of first-type transistors includes the fabrication of the gate-conductors intersecting the first semiconductor structure at channel regions of the first-type transistors. The fabrication of second-type transistors includes the fabrication of the gate-conductors intersecting the second semiconductor structure at channel regions of the second-type transistors. In some embodiments, the fabrication of first-type transistors also includes the fabrication of the terminal-conductors intersecting the first semiconductor structure at source/drain regions of the first-type transistors. In some embodiments, the fabrication of second-type transistors also includes the fabrication of the terminal-conductors intersecting the second semiconductor structure at source/drain regions of the second-type transistors. In the non-limiting examples of FIG. 2A and FIG. 2B, the terminal-conductors 172p, 172n, and 178 are fabricated in operation 1120 when the p-type transistor T1p and the n-type transistor T1n are fabricated. In FIG. 2A, the terminal-conductor 172p intersects the semiconductor structure 180p at the source region sT1p of the p-type transistor T1p, and the terminal-conductor 172n intersects the n-type first semiconductor structure 180n at the source region sT1n of the n-type transistor T1n. In FIG. 2B, the terminal-conductor 178 intersects the semiconductor structure 180p at the drain region dT1p of the p-type transistor T1p

and intersects the semiconductor structure **180n** at the drain region dT**1n** of the n-type transistor T**1n**.

[0082] In operation **1130** of method **1100**, the first-voltage underlayer power rail and the second-voltage underlayer power rail are fabricated. The first-voltage underlayer power rail is conductively connecting to the source region of the first-type transistor. The second-voltage underlayer power rail is conductively connecting to the source region of the second-type transistor. In the non-limiting examples of FIGS. 2A-2C, the first-voltage underlayer power rail **40** and the second-voltage underlayer power rail **20** are fabricated above the semiconductor structure **180p** and the semiconductor structure **180n**. In FIG. 2A, the first-voltage underlayer power rail **40** is conductively connected to the terminal-conductor **172p**, and the second-voltage underlayer power rail **20** is conductively connected to the terminal-conductor **172n**.

[0083] In operation **1140** of method **1100**, an insulating material is deposited covering the first-voltage underlayer power rail, the second-voltage underlayer power rail. In the non-limiting examples of FIGS. 2A-2C, the insulating material **210** are deposited covering the first-voltage underlayer power rail **40** and the second-voltage underlayer power rail **20**.

[0084] In operation **1150** of method **1100**, metals are deposited over the insulating material to form the first connection layer. Then, in operation **1155** of method **1100**, the first connection layer is patterned to form the first-voltage power rail, the second-voltage power rail, the first signal conducting line, and the second signal conducting line. The first-voltage power rail is directly connected with the first-voltage underlayer power rail through a first via-connector and the second-voltage power rail is directly connected with the second-voltage underlayer power rail through a second via-connector. The first signal conducting line is directly connected with the gate-conductor through a gate via-connector. In the non-limiting examples of FIGS. 2A-2C, metals are deposited over the insulating material **210** to form the first connection layer M**0**. The signal conducting line **198** in FIG. 2B, the first-voltage power rail **58** in FIG. 2C, and the second-voltage power rail **38** in FIG. 2C are all formed in operation **1155** by patterning the first connection layer M**0**. In FIG. 2B, the signal conducting line **198** is conductively connected to the terminal-conductor **178** through the terminal via-connector VD. In FIG. 2C, The first-voltage power rail **58** is directly connected with the first-voltage underlayer power rail **40** through the via-connector V**58**, and the second-voltage power rail **38** is directly connected with the second-voltage underlayer power rail **20** through the via-connector V**38**.

[0085] In operation **1160** of method **1100**, metals are deposited over a layer of inter-layer dielectric to form a second connection layer. In operation **1165** of method **1100**, the second connection layer are patterned to form the first-voltage power stub and the second-voltage power stub. In the non-limiting examples of FIG. 8, the first-voltage power rails (**852** and **854**) and the second-voltage power rail **832** are all formed in operation **1155** by patterning the first connection layer M**0**. Each of the first-voltage power rails (**852** and **854**) is directly connected with the first-voltage underlayer power rail **40A**. The second-voltage power rail **832** is directly connected with the second-voltage underlayer power rails **20A** and **20B**. The first-voltage power stubs (S**852** and S**854**) and the second-voltage power stub (S**832**) are all formed in operation **1165** by patterning the second connection layer M**1**. The second connection layer M**1** is above the first connection layer M**0** and separated from interlayer dielectric (ILD) materials (not shown in FIG. 8).

[0086] FIG. 12 is a block diagram of an electronic design automation (EDA) system **1200** in accordance with some embodiments.

[0087] In some embodiments, EDA system **1200** includes an APR system. Methods described herein of designing layout diagrams represent wire routing arrangements, in accordance with one or more embodiments, are implementable, for example, using EDA system **1200**, in accordance with some embodiments.

[0088] In some embodiments, EDA system **1200** is a general purpose computing device including a hardware processor **1202** and a non-transitory, computer-readable storage medium **1204**. Storage

medium **1204**, amongst other things, is encoded with, i.e., stores, computer program code **1206**, i.e., a set of executable instructions. Execution of instructions **1206** by hardware processor **1202** represents (at least in part) an EDA tool which implements a portion or all of the methods described herein in accordance with one or more embodiments (hereinafter, the noted processes and/or methods).

[0089] Processor **1202** is electrically coupled to computer-readable storage medium **1204** via a bus **1208**. Processor **1202** is also electrically coupled to an I/O interface **1210** by bus **1208**. A network interface **1212** is also electrically connected to processor **1202** via bus **1208**. Network interface **1212** is connected to a network **1214**, so that processor **1202** and computer-readable storage medium **1204** are capable of connecting to external elements via network **1214**. Processor **1202** is configured to execute computer program code **1206** encoded in computer-readable storage medium **1204** in order to cause system **1200** to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, processor **1202** is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

[0090] In one or more embodiments, computer-readable storage medium **1204** is an electronic, magnetic, optical, electromagnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, computer-readable storage medium **1204** includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In one or more embodiments using optical disks, computer-readable storage medium **1204** includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

[0091] In one or more embodiments, storage medium **1204** stores computer program code **1206** configured to cause system **1200** (where such execution represents (at least in part) the EDA tool) to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, storage medium **1204** also stores information which facilitates performing a portion or all of the noted processes and/or methods. In one or more embodiments, storage medium **1204** stores library **1207** of standard cells including such standard cells as disclosed herein. In one or more embodiments, storage medium **1204** stores one or more layout diagrams **1209** corresponding to one or more layouts disclosed herein.

[0092] EDA system **1200** includes I/O interface **1210**. I/O interface **1210** is coupled to external circuitry. In one or more embodiments, I/O interface **1210** includes a keyboard, keypad, mouse, trackball, trackpad, touchscreen, and/or cursor direction keys for communicating information and commands to processor **1202**.

[0093] EDA system **1200** also includes network interface **1212** coupled to processor **1202**. Network interface **1212** allows system **1200** to communicate with network **1214**, to which one or more other computer systems are connected. Network interface **1212** includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interfaces such as ETHERNET, USB, or IEEE-1364. In one or more embodiments, a portion or all of noted processes and/or methods, is implemented in two or more systems **1200**.

[0094] System **1200** is configured to receive information through I/O interface **1210**. The information received through I/O interface **1210** includes one or more of instructions, data, design rules, libraries of standard cells, and/or other parameters for processing by processor **1202**. The information is transferred to processor **1202** via bus **1208**. EDA system **1200** is configured to receive information related to a UI through I/O interface **1210**. The information is stored in computer-readable medium **1204** as user interface (UI) **1242**.

[0095] In some embodiments, a portion or all of the noted processes and/or methods is implemented as a standalone software application for execution by a processor. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software

application that is a part of an additional software application. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a plug-in to a software application. In some embodiments, at least one of the noted processes and/or methods is implemented as a software application that is a portion of an EDA tool. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is used by EDA system **1200**. In some embodiments, a layout diagram which includes standard cells is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool.

[0096] In some embodiments, the processes are realized as functions of a program stored in a non-transitory computer readable recording medium. Examples of a non-transitory computer readable recording medium include, but are not limited to, external/removable and/or internal/built-in storage or memory unit, e.g., one or more of an optical disk, such as a DVD, a magnetic disk, such as a hard disk, a semiconductor memory, such as a ROM, a RAM, a memory card, and the like.

[0097] FIG. **13** is a block diagram of an integrated circuit (IC) manufacturing system **1300**, and an IC manufacturing flow associated therewith, in accordance with some embodiments. In some embodiments, based on a layout diagram, at least one of (A) one or more semiconductor masks or (B) at least one component in a layer of a semiconductor integrated circuit is fabricated using manufacturing system **1300**.

[0098] In FIG. **13**, IC manufacturing system **1300** includes entities, such as a design house **1320**, a mask house **1330**, and an IC manufacturer/fabricator (“fab”) **1350**, that interact with one another in the design, development, and manufacturing cycles and/or services related to manufacturing an IC device **1360**. The entities in system **1300** are connected by a communications network. In some embodiments, the communications network is a single network. In some embodiments, the communications network is a variety of different networks, such as an intranet and the Internet. The communications network includes wired and/or wireless communication channels. Each entity interacts with one or more of the other entities and provides services to and/or receives services from one or more of the other entities. In some embodiments, two or more of design house **1320**, mask house **1330**, and IC fab **1350** is owned by a single larger company. In some embodiments, two or more of design house **1320**, mask house **1330**, and IC fab **1350** coexist in a common facility and use common resources.

[0099] Design house (or design team) **1320** generates an IC design layout diagram **1322**. IC design layout diagram **1322** includes various geometrical patterns designed for an IC device **1360**. The geometrical patterns correspond to patterns of metal, oxide, or semiconductor layers that make up the various components of IC device **1360** to be fabricated. The various layers combine to form various IC features. For example, a portion of IC design layout diagram **1322** includes various IC features, such as an active region, gate electrode, source and drain, metal lines or vias of an interlayer interconnection, and openings for bonding pads, to be formed in a semiconductor substrate (such as a silicon wafer) and various material layers disposed on the semiconductor substrate. Design house **1320** implements a proper design procedure to form IC design layout diagram **1322**. The design procedure includes one or more of logic design, physical design or place and route. IC design layout diagram **1322** is presented in one or more data files having information of the geometrical patterns. For example, IC design layout diagram **1322** can be expressed in a GDSII file format or DFII file format.

[0100] Mask house **1330** includes data preparation **1332** and mask fabrication **1344**. Mask house **1330** uses IC design layout diagram **1322** to manufacture one or more masks **1345** to be used for fabricating the various layers of IC device **1360** according to IC design layout diagram **1322**. Mask house **1330** performs mask data preparation **1332**, where IC design layout diagram **1322** is translated into a representative data file (“RDF”). Mask data preparation **1332** provides the RDF to mask fabrication **1344**. Mask fabrication **1344** includes a mask writer. A mask writer converts the RDF to an image on a substrate, such as a mask (reticle) **1345** or a semiconductor wafer **1353**. The

design layout diagram **1322** is manipulated by mask data preparation **1332** to comply with particular characteristics of the mask writer and/or requirements of IC fab **1350**. In FIG. **13**, mask data preparation **1332** and mask fabrication **1344** are illustrated as separate elements. In some embodiments, mask data preparation **1332** and mask fabrication **1344** can be collectively referred to as mask data preparation.

[0101] In some embodiments, mask data preparation **1332** includes optical proximity correction (OPC) which uses lithography enhancement techniques to compensate for image errors, such as those that can arise from diffraction, interference, other process effects and the like. OPC adjusts IC design layout diagram **1322**. In some embodiments, mask data preparation **1332** includes further resolution enhancement techniques (RET), such as off-axis illumination, sub-resolution assist features, phase-shifting masks, other suitable techniques, and the like or combinations thereof. In some embodiments, inverse lithography technology (ILT) is also used, which treats OPC as an inverse imaging problem.

[0102] In some embodiments, mask data preparation **1332** includes a mask rule checker (MRC) that checks the IC design layout diagram **1322** that has undergone processes in OPC with a set of mask creation rules which contain certain geometric and/or connectivity restrictions to ensure sufficient margins, to account for variability in semiconductor manufacturing processes, and the like. In some embodiments, the MRC modifies the IC design layout diagram **1322** to compensate for limitations during mask fabrication **1344**, which may undo part of the modifications performed by OPC in order to meet mask creation rules.

[0103] In some embodiments, mask data preparation **1332** includes lithography process checking (LPC) that simulates processing that will be implemented by IC fab **1350** to fabricate IC device **1360**. LPC simulates this processing based on IC design layout diagram **1322** to create a simulated manufactured device, such as IC device **1360**. The processing parameters in LPC simulation can include parameters associated with various processes of the IC manufacturing cycle, parameters associated with tools used for manufacturing the IC, and/or other aspects of the manufacturing process. LPC takes into account various factors, such as aerial image contrast, depth of focus (“DOF”), mask error enhancement factor (“MEEF”), other suitable factors, and the like or combinations thereof. In some embodiments, after a simulated manufactured device has been created by LPC, if the simulated device is not close enough in shape to satisfy design rules, OPC and/or MRC are repeated to further refine IC design layout diagram **1322**.

[0104] It should be understood that the above description of mask data preparation **1332** has been simplified for the purposes of clarity. In some embodiments, data preparation **1332** includes additional features such as a logic operation (LOP) to modify the IC design layout diagram **1322** according to manufacturing rules. Additionally, the processes applied to IC design layout diagram **1322** during data preparation **1332** may be executed in a variety of different orders.

[0105] After mask data preparation **1332** and during mask fabrication **1344**, a mask **1345** or a group of masks **1345** are fabricated based on the modified IC design layout diagram **1322**. In some embodiments, mask fabrication **1344** includes performing one or more lithographic exposures based on IC design layout diagram **1322**. In some embodiments, an electron-beam (e-beam) or a mechanism of multiple e-beams is used to form a pattern on a mask (photomask or reticle) **1345** based on the modified IC design layout diagram **1322**. Mask **1345** can be formed in various technologies. In some embodiments, mask **1345** is formed using binary technology. In some embodiments, a mask pattern includes opaque regions and transparent regions. A radiation beam, such as an ultraviolet (UV) beam, used to expose the image sensitive material layer (e.g., photoresist) which has been coated on a wafer, is blocked by the opaque region and transmits through the transparent regions. In one example, a binary mask version of mask **1345** includes a transparent substrate (e.g., fused quartz) and an opaque material (e.g., chromium) coated in the opaque regions of the binary mask. In another example, mask **1345** is formed using a phase shift technology. In a phase shift mask (PSM) version of mask **1345**, various features in the pattern

formed on the phase shift mask are configured to have proper phase difference to enhance the resolution and imaging quality. In various examples, the phase shift mask can be attenuated PSM or alternating PSM. The mask(s) generated by mask fabrication **1344** is used in a variety of processes. For example, such a mask(s) is used in an ion implantation process to form various doped regions in semiconductor wafer **1353**, in an etching process to form various etching regions in semiconductor wafer **1353**, and/or in other suitable processes.

[0106] IC fab **1350** is an IC fabrication business that includes one or more manufacturing facilities for the fabrication of a variety of different IC products. In some embodiments, IC Fab **1350** is a semiconductor foundry. For example, there may be a manufacturing facility for the front end fabrication of a plurality of IC products (front-end-of-line (FEOL) fabrication), while a second manufacturing facility may provide the back end fabrication for the interconnection and packaging of the IC products (back-end-of-line (BEOL) fabrication), and a third manufacturing facility may provide other services for the foundry business.

[0107] IC fab **1350** includes fabrication tools **1352** configured to execute various manufacturing operations on semiconductor wafer **1353** such that IC device **1360** is fabricated in accordance with the mask(s), e.g., mask **1345**. In various embodiments, fabrication tools **1352** include one or more of a wafer stepper, an ion implanter, a photoresist coater, a process chamber, e.g., a CVD chamber or LPCVD furnace, a CMP system, a plasma etch system, a wafer cleaning system, or other manufacturing equipment capable of performing one or more suitable manufacturing processes as discussed herein.

[0108] IC fab **1350** uses mask(s) **1345** fabricated by mask house **1330** to fabricate IC device **1360**. Thus, IC fab **1350** at least indirectly uses IC design layout diagram **1322** to fabricate IC device **1360**. In some embodiments, semiconductor wafer **1353** is fabricated by IC fab **1350** using mask(s) **1345** to form IC device **1360**. In some embodiments, the IC fabrication includes performing one or more lithographic exposures based at least indirectly on IC design layout diagram **1322**.

Semiconductor wafer **1353** includes a silicon substrate or other proper substrate having material layers formed thereon. Semiconductor wafer **1353** further includes one or more of various doped regions, dielectric features, multilevel interconnects, and the like (formed at subsequent manufacturing steps).

[0109] Details regarding an integrated circuit (IC) manufacturing system (e.g., system **1300** of FIG. **13**), and an IC manufacturing flow associated therewith are found, e.g., in U.S. Pat. No. 9,256,709, granted Feb. 9, 2016, U.S. Pre-Grant Publication No. 20150278429, published Oct. 1, 2015, U.S. Pre-Grant Publication No. 20140040838, published Feb. 6, 2014, and U.S. Pat. No. 7,260,442, granted Aug. 21, 2007, the entireties of each of which are hereby incorporated by reference.

[0110] An aspect of the present disclosure relates to an integrated circuit. The integrated circuit includes a first-voltage power rail and a second-voltage power rail, each of the first-voltage power rail and the second-voltage power rail extending in a second direction that is perpendicular a first direction, each of the first-voltage power rail and the second-voltage power rail being in a first connection layer; a first-voltage underlayer power rail and a second-voltage underlayer power rail, each of the first-voltage underlayer power rail and the second-voltage underlayer power rail extending in the first direction, each of the first-voltage underlayer power rail and the second-voltage underlayer power rail being below the first connection layer, where the first-voltage underlayer power rail is conductively connected to the first-voltage power rail, and the second-voltage underlayer power rail is conductively connected to the second-voltage power rail; a first-type transistor having a source region, in a first semiconductor structure within a first-type active zone underneath the first connection layer, conductively connected to the first-voltage underlayer power rail; a second-type transistor having a source region, in a second semiconductor structure within a second-type active zone underneath the first connection layer, conductively connected to the second-voltage underlayer power rail; a first signal conducting line and a second signal conducting line extending in the second direction, each of the first signal conducting line and the

second signal conducting line being in the first connection layer; a terminal-conductor intersecting the first semiconductor structure at a drain region of the first-type transistor or intersecting the second semiconductor structure at a drain region of the second-type transistor; and a terminal via-connector directly connecting the terminal-conductor with the second signal conducting line.

[0111] Another aspect of the present disclosure relates to an integrated circuit. The integrated circuit includes a first-type transistor having a source region in a first semiconductor structure extending in a first direction within a first-type active zone; a first-voltage power rail extending in a second direction in a first connection layer, where the second direction is perpendicular the first direction; a first-voltage underlayer power rail extending in the first direction underneath the first connection layer, where the first-voltage underlayer power rail is conductively connected to the source region of the first-type transistor and the first-voltage power rail; a second-type transistor having a source region in a second semiconductor structure extending in the first direction within a second-type active zone; a second-voltage power rail extending in the second direction in the first connection layer; a second-voltage underlayer power rail extending in the first direction underneath the first connection layer, where the second-voltage underlayer power rail is conductively connected to the source region of the second-type transistor and the second-voltage power rail; a first signal conducting line and a second signal conducting line extending in the second direction, each of the first signal conducting line and the second signal conducting line being in the first connection layer; a terminal-conductor intersecting the first semiconductor structure at a drain region of the first-type transistor or intersecting the second semiconductor structure at a drain region of the second-type transistor; and a terminal via-connector directly connecting the terminal-conductor with the second signal conducting line.

[0112] Still another aspect of the present disclosure relates to an integrated circuit. The integrated circuit includes a first-type transistor having a source region in a first semiconductor structure extending in a first direction within a first-type active zone; a first-voltage power rail extending in a second direction in a first connection layer, where the second direction is perpendicular the first direction; a first-voltage underlayer power rail extending in the first direction underneath the first connection layer, where the first-voltage underlayer power rail is conductively connected to the source region of the first-type transistor and the first-voltage power rail; a second-type transistor having a source region in a second semiconductor structure extending in the first direction within a second-type active zone; a second-voltage power rail extending in the second direction in the first connection layer; a second-voltage underlayer power rail extending in the first direction underneath the first connection layer, where the second-voltage underlayer power rail is conductively connected to the source region of the second-type transistor and the second-voltage power rail; a first signal conducting line and a second signal conducting line extending in the second direction, each of the first signal conducting line and the second signal conducting line being in the first connection layer; a gate-conductor extending in the second direction, where the gate-conductor intersects either the first semiconductor structure or the second semiconductor structure at a channel region; and a gate via-connector directly connecting the gate-conductor with the first signal conducting line.

[0113] It will be readily seen by one of ordinary skill in the art that one or more of the disclosed embodiments fulfill one or more of the advantages set forth above. After reading the foregoing specification, one of ordinary skill will be able to affect various changes, substitutions of equivalents and various other embodiments as broadly disclosed herein. It is therefore intended that the protection granted hereon be limited only by the definition contained in the appended claims and equivalents thereof.

Claims

- 1.** An integrated circuit comprising: a first-voltage power rail and a second-voltage power rail, each of the first-voltage power rail and the second-voltage power rail extending in a second direction that is perpendicular a first direction, each of the first-voltage power rail and the second-voltage power rail being in a first connection layer; a first-voltage underlayer power rail and a second-voltage underlayer power rail, each of the first-voltage underlayer power rail and the second-voltage underlayer power rail extending in the first direction, each of the first-voltage underlayer power rail and the second-voltage underlayer power rail being below the first connection layer, wherein the first-voltage underlayer power rail is conductively connected to the first-voltage power rail, and the second-voltage underlayer power rail is conductively connected to the second-voltage power rail; a first-type transistor having a source region, in a first semiconductor structure within a first-type active zone underneath the first connection layer, conductively connected to the first-voltage underlayer power rail; a second-type transistor having a source region, in a second semiconductor structure within a second-type active zone underneath the first connection layer, conductively connected to the second-voltage underlayer power rail; a first signal conducting line and a second signal conducting line extending in the second direction, each of the first signal conducting line and the second signal conducting line being in the first connection layer; a terminal-conductor intersecting the first semiconductor structure at a drain region of the first-type transistor or intersecting the second semiconductor structure at a drain region of the second-type transistor; and a terminal via-connector directly connecting the terminal-conductor with the second signal conducting line.
- 2.** The integrated circuit of claim 1, further comprising: a gate-conductor extending in the second direction, wherein the gate-conductor intersects either the first semiconductor structure or the second semiconductor structure at a channel region.
- 3.** The integrated circuit of claim 2, further comprising: a gate via-connector directly connecting the gate-conductor with the first signal conducting line.
- 4.** The integrated circuit of claim 1, further comprising: a first-voltage power stub and a second-voltage power stub, each of the first-voltage power stub and the second-voltage power stub extending in the first direction, each of the first-voltage power stub and the second-voltage power stub being in a second connection layer above the first connection layer.
- 5.** The integrated circuit of claim 4, wherein the first-voltage power stub is conductively connected to the first-voltage power rail and the second-voltage power stub is conductively connected to the second-voltage power rail.
- 6.** The integrated circuit of claim 5, further comprising: a first-voltage third-connection power stub extending in the second direction in a third connection layer above the second connection layer; and wherein the first-voltage third-connection power stub is conductively connected to the first-voltage power stub.
- 7.** The integrated circuit of claim 6, further comprising: a first-voltage fourth-connection power stub extending in the first direction in a fourth connection layer above the second connection layer; and wherein the first-voltage fourth-connection power stub is conductively connected to the first-voltage third-connection power stub.
- 8.** An integrated circuit comprising: a first-type transistor having a source region in a first semiconductor structure extending in a first direction within a first-type active zone; a first-voltage power rail extending in a second direction in a first connection layer, wherein the second direction is perpendicular the first direction; a first-voltage underlayer power rail extending in the first direction underneath the first connection layer, wherein the first-voltage underlayer power rail is conductively connected to the source region of the first-type transistor and the first-voltage power rail; a second-type transistor having a source region in a second semiconductor structure extending in the first direction within a second-type active zone; a second-voltage power rail extending in the second direction in the first connection layer; a second-voltage underlayer power rail extending in

the first direction underneath the first connection layer, wherein the second-voltage underlayer power rail is conductively connected to the source region of the second-type transistor and the second-voltage power rail; a first signal conducting line and a second signal conducting line extending in the second direction, each of the first signal conducting line and the second signal conducting line being in the first connection layer; a terminal-conductor intersecting the first semiconductor structure at a drain region of the first-type transistor or intersecting the second semiconductor structure at a drain region of the second-type transistor; and a terminal via-connector directly connecting the terminal-conductor with the second signal conducting line.

9. The integrated circuit of claim 8, further comprising: a gate-conductor extending in the second direction, wherein the gate-conductor intersects either the first semiconductor structure or the second semiconductor structure at a channel region.

10. The integrated circuit of claim 9, further comprising: a gate via-connector directly connecting the gate-conductor with the first signal conducting line.

11. The integrated circuit of claim 8, further comprising: a first-voltage power stub and a second-voltage power stub, each of the first-voltage power stub and the second-voltage power stub extending in the first direction, each of the first-voltage power stub and the second-voltage power stub being in a second connection layer above the first connection layer.

12. The integrated circuit of claim 11, wherein the first-voltage power stub is conductively connected to the first-voltage power rail and the second-voltage power stub is conductively connected to the second-voltage power rail.

13. The integrated circuit of claim 12, further comprising: a first-voltage third-connection power stub extending in the second direction in a third connection layer above the second connection layer; and wherein the first-voltage third-connection power stub is conductively connected to the first-voltage power stub.

14. The integrated circuit of claim 13, further comprising: a first-voltage fourth-connection power stub extending in the first direction in a fourth connection layer above the second connection layer; and wherein the first-voltage fourth-connection power stub is conductively connected to the first-voltage third-connection power stub.

15. An integrated circuit comprising: a first-type transistor having a source region in a first semiconductor structure extending in a first direction within a first-type active zone; a first-voltage power rail extending in a second direction in a first connection layer, wherein the second direction is perpendicular the first direction; a first-voltage underlayer power rail extending in the first direction underneath the first connection layer, wherein the first-voltage underlayer power rail is conductively connected to the source region of the first-type transistor and the first-voltage power rail; a second-type transistor having a source region in a second semiconductor structure extending in the first direction within a second-type active zone; a second-voltage power rail extending in the second direction in the first connection layer; a second-voltage underlayer power rail extending in the first direction underneath the first connection layer, wherein the second-voltage underlayer power rail is conductively connected to the source region of the second-type transistor and the second-voltage power rail; a first signal conducting line and a second signal conducting line extending in the second direction, each of the first signal conducting line and the second signal conducting line being in the first connection layer; a gate-conductor extending in the second direction, wherein the gate-conductor intersects either the first semiconductor structure or the second semiconductor structure at a channel region; and a gate via-connector directly connecting the gate-conductor with the first signal conducting line.

16. The integrated circuit of claim 15, further comprising: a terminal-conductor intersecting the first semiconductor structure at a drain region of the first-type transistor or intersecting the second semiconductor structure at a drain region of the second-type transistor.

17. The integrated circuit of claim 16, further comprising: a terminal via-connector directly connecting the terminal-conductor with the second signal conducting line.

- 18.** The integrated circuit of claim 15, further comprising: a first-voltage power stub and a second-voltage power stub, each of the first-voltage power stub and the second-voltage power stub extending in the first direction, each of the first-voltage power stub and the second-voltage power stub being in a second connection layer above the first connection layer.
- 19.** The integrated circuit of claim 18, wherein the first-voltage power stub is conductively connected to the first-voltage power rail and the second-voltage power stub is conductively connected to the second-voltage power rail.
- 20.** The integrated circuit of claim 19, further comprising: a first-voltage third-connection power stub extending in the second direction in a third connection layer above the second connection layer; and wherein the first-voltage third-connection power stub is conductively connected to the first-voltage power stub.
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